Project Proposal Template

Project: [Federal Workforce Intelligence & Retention Analytics]

Objective:. To analyze large-scale U.S. federal workforce data to understand workforce size, hiring and separation trends, employee movement, workforce composition, and retention patterns. The project aims to provide data-driven insights that can support workforce planning, resource allocation, and retention strategies.

Description: The project will use large-scale Federal Workforce Data from the U.S. Office of Personnel Management (OPM), including Federal Employment, Federal Accessions, and Federal Separations datasets. The raw data is provided in large TXT files and contains millions of recorThe project will process and transform the raw data using Python and SQL Server, build a structured data warehouse using Bronze, Silver, and Gold layers, and create interactive dashboards using Power BI/Tableau. The final solution will provide KPIs and analytical insights related to workforce trends, accessions, separations, and workforce planning.ds.

Tools & Design: - Data Source: U.S. Office of Personnel Management (OPM)
- Raw Data Format: TXT
- Data Processing: Python
- Database: SQL Server
- Data Warehouse Architecture: Bronze → Silver → Gold
- Visualization: Power BI / Tableau
- Version Control: Git & GitHub
- Data Modeling: Star Schema
- Analytics: Workforce trends, accessions, separations, retention and workforce planning

Technologies to Use: - Python
- Pandas
- NumPy
- SQL
- SQL Server
- Power BI

- Tableau
- Git & GitHub
- Jupyter Notebook

Team Members Name

1- **Gaser Mohammed Naiem (Team Leader)**
2- **Hana Faisal Ezzat**
3- **Mohammed Hossam Khamis**
4- **Yara Mohammed Mostafa**
5- **Mostafa Samer Ali**

Team Members & Responsibilities:

1- Team Member 1 – Project Lead & Business Analyst
- Define the business problem and objectives.
- Analyze the OPM Data Dictionary.
- Define KPIs and business questions.
- Coordinate the team and track project progress.

2- Team Member 2 – Data Engineer / Data Collection
- Download and organize the raw TXT datasets.
- Inspect file structures and delimiters.
- Handle large-scale data ingestion.
- Prepare the raw data for processing.

3- Team Member 3 – Python Data Analyst
- Profile the datasets using Python.
- Handle missing values and duplicates.
- Clean and standardize data.
- Perform Exploratory Data Analysis (EDA).

4- Team Member 4 – SQL / Data Warehouse Engineer
- Design the database structure.
- Build Bronze, Silver, and Gold layers.

- Create fact and dimension tables.
- Develop SQL transformations and analytical queries.

5- Team Member 5 – BI Developer
- Connect the Gold layer to Power BI/Tableau.
- Build interactive dashboards.
- Create KPIs, filters, charts, and drill-downs.
- Develop workforce and retention visualizations.

Week 1: *[Data Understanding, Collection & Profiling]*

- **Member 1:** Define the business problem, objectives, business questions, and project KPIs.
- **Member 2:** Download and organize the Employment, Accessions, and Separations TXT files.
- **Member 3:** Inspect the TXT files using Python and identify columns, data types, null values, and duplicates.
- **Member 4:** Study the OPM Data Dictionary and identify potential keys and relationships.
- **Member 5:** Define the required dashboard pages and visualization requirements.

Deliverables:

- Project requirements and business questions.
- Raw TXT datasets organized by source.
- OPM Data Dictionary analysis.
- Initial data profiling report.
- Initial KPI list.
- Initial dashboard design/wireframe

Week 2: *[Data Cleaning, Transformation & Data Warehouse]*

Member 1: Validate the business rules and KPI definitions.
- **Member 2:** Develop the Python data ingestion process for the large TXT files.
- **Member 3:** Clean missing values, duplicates, inconsistent formats, and categorical fields.
- **Member 4:** Design and implement the SQL Server Data Warehouse.
- **Member 5:** Prepare the BI data requirements and define required measures.

Deliverables:

- Cleaned and standardized datasets.
- Python data-processing scripts/notebooks.
- SQL Server database structure.
- Bronze and Silver layers.
- Initial exploratory analysis.
- Updated data model and KPI definitions

Week 3: *[Gold Layer, Analytics & Dashboard Development]*

- **Member 1:** Validate analytical requirements and business logic.
- **Member 2:** Support the ETL process and large-scale data loading.
- **Member 3:** Calculate workforce, accession, and separation metrics using Python/SQL.
- **Member 4:** Build the Gold layer and fact/dimension tables.
- **Member 5:** Develop the main Power BI/Tableau dashboards.

Deliverables:

- Completed Gold layer.
- Finalized Star Schema.
- Workforce KPI calculations.
- First dashboard version.
- Accessions and Separations analysis.
- Initial business insights and analytical findings.

Week 4: *[Dashboard Finalization, Validation & Presentation]*

- **Member 1:** Review the complete project and validate that business questions are answered.
- **Member 2:** Validate the final data pipeline and ensure large TXT files are processed correctly.
- **Member 3:** Perform final data-quality checks and validate analytical results.
- **Member 4:** Optimize SQL queries and finalize the Data Warehouse.
- **Member 5:** Finalize dashboards, interactions, filters, KPIs, and visual design.

Deliverables:

- Final validated Data Warehouse.
- Final interactive Power BI/Tableau dashboards.
- Data-quality and validation report.
- Business insights and recommendations.
- Complete project documentation.
- Final presentation and project demonstration.